Assignment Alpha

Course Artificial Agentic AI	Course Code A4051
Anncouement date 16-02-2026	Due Date With in 15 working days after posting of the assignment on a GCR 2nd March 2026
Total marks 100	Section A and B

Instructions:

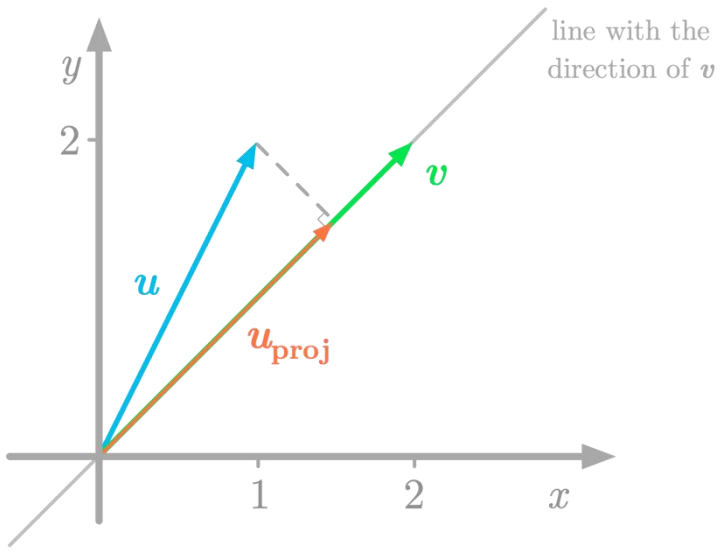
1- Submit in soft form only. In case you have handwritten notes, scan them and upload properly.
2- Don't copy other's solutions.
3- Your content should be AI free.
4- There will be random Viva as well. Hence, must be sure what you have written in your solution.
5- For any ambiguity, you may assume values.

Question no 1 **Marks [15]**

1. In a word embedding vector space, If A=(6,−1,√9), for what value of **X** is the vector b=(4,X,−2) perpendicular to A?

2. What is the difference between dot product and Hadamard product between two vectors?

3. Consider the following figure. Prove from vector project and L norm that the dot product between u and v is 6.



Question no 2 **Marks [10]**

Calculate the **Attention Score** of the following sentence which is represented by 2-dimension vector.

Agentic AI emerging course

The following vectors are given:

Agentic Q = [1.5, 3], Agentic V= [0.2,0.3] Agentic K=[1.1, 2.2]

AI Q = [2, 4], AI V= [2,0.3] AI K=[1, 3]

emerging Q = [2.2, 4.4], emerging V= [2.2,0.3] emerging K=[1, 3]

course Q = [2.4, 4.4], course V= [2.2,0.3] course K=[1.1, 3]

Question no 3 **Marks [16]**

Consider the following figure, where a **learning agent (drone)** is used to deliver a parcel box in the city. Before delivery, the drone agent learns about the weather and predicts rain chances based on wind and humidity. It delivers the parcel if there is no prediction /chances of rain. **Determine the following:**

1-Learning agent uses Naive Bayesian learning algorithm to predict rain for parcel delivery. Consider the agent uses Laplacian smoothing to avoid zero probabilities.

2-Learning agent uses information gain and entropy learning parameters to predict rain for parcel delivery.

3-Learning agent uses logistic regression learning algorithm to predict rain for parcel delivery.

4-Compare all the learning algorithm used by the learning agent.

Show all calculations and develop a notebook of your solution as well.

Question no 4 **Marks [15]**

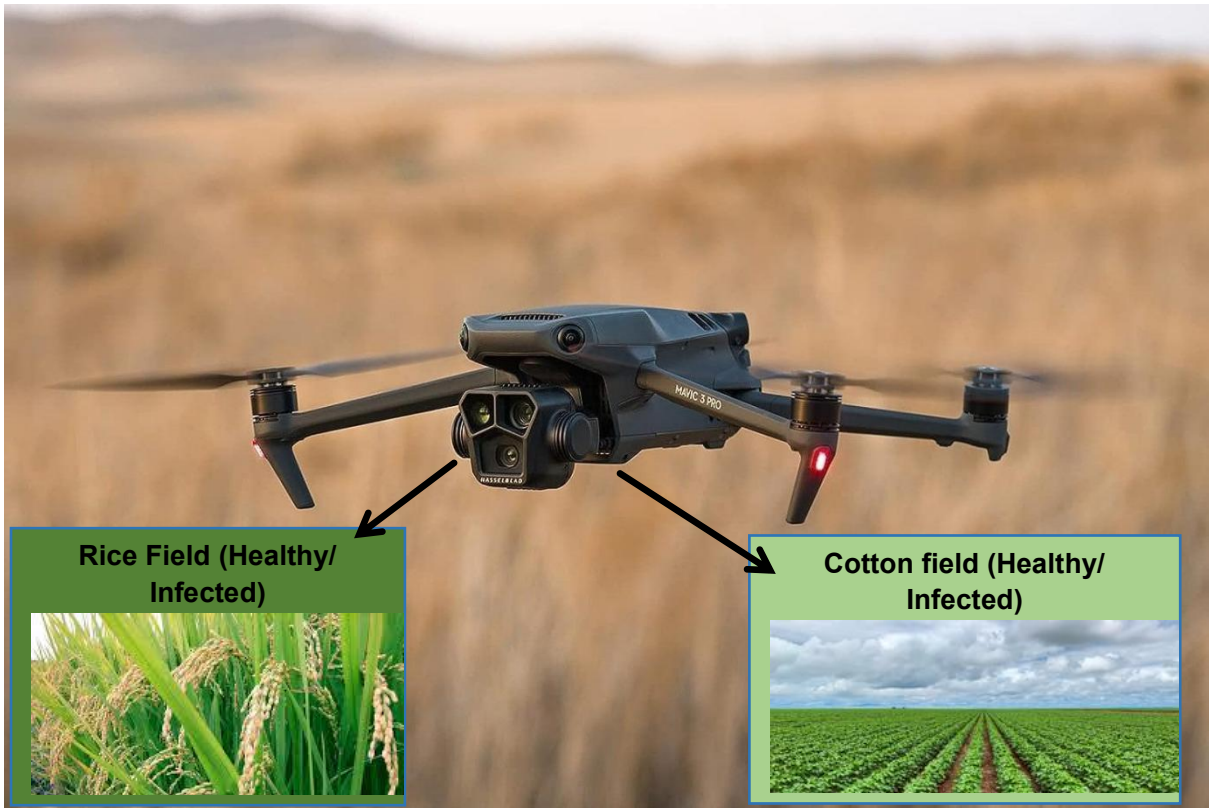
Develop **Eco-Friendly agents** that emits less **Co_2** and water, for the following information.

A- Reflex agent for cleaning wastage on street.
B- Utility-based agent for earning flying hours with the airline.
C- Goal-based agent in in-drive taxi to provide secure commute.

P. S You can use any coding platform that you are comfortable with. Provide a complete notebook along with detailed description.

Question no 5 **Marks [6+6]**

Consider the following scenario as shown in the figure below, where a drone is being used to monitor a rice field **(location A)** and a cotton field **(location B)**. The drone is monitoring crop condition that is classified as healthy or infected. In case an infected crop is detected by the drone, it starts spraying medicines.



The drone is acting as a reflex agent to execute the tasks based on the visual input it is perceiving at time stamps t=0, 1, 2 and 3. The drone receives credits worth $2 for a healthy field at each time stamp. If the farm field is infected, the drone starts spraying; else, if it is healthy, it moves to the next field. We assume that after the spray process,

the field becomes healthy and may get infected after 10 time stamps. Assume that initial disease spreading in both fields are independent with equal probability.

I. Determine the joint probability of both field being infected.
II. Assume that initial state of the field are (A= Infected, B = Infected) and drone is monitoring location A currently and will move to next location after performing some task. Determine credits that drone earns for time t=0 ,1,2 and 3. Develop a precept and action table at time t=0,1,2 and 3.

Question no 6 **Marks [12]**

Complete the following pseud-code and extend it for the learning agent. Also, write the python code.

```
function multi_head_attention(Q, ???, V, num_heads, d_model, mask=None):
    '''
    Q, K, V: (batch, seq_len, d_model)
    num_heads: number of attention heads
    d_model: dimension of the model (must be divisible by num_heads)
    '''
    batch_size = Q.shape[???]
    d_k = d_model / num_heads  # dimension per head

    # Linear projections for Q, K, V
    Q_proj = Q @ W_Q  # (batch, seq_len, ???)
    K_proj = K @ W_K  # (batch, seq_len, d_model)
    V_proj = V @ W_V  # (batch, seq_len, d_model)
    # (batch, seq_len, d_model) => (batch, seq_len, num_heads, d_k)
    # => (batch, num_heads, seq_len, d_k)
    Q_heads = split_heads(Q_proj, num_heads)
    K_heads = split_heads(K_proj, num_heads)
    V_heads = split_heads(???, num_heads)

    attention_output, attention_weights = scaled_dot_product_attention(
        Q_heads, K_heads, V_heads, mask
    )
    # Output shape: (batch, num_heads, seq_len, d_k)

    # Concatenate heads
    # (batch, num_heads, seq_len, d_k) => (batch, seq_len, d_model)
    attention_output = concatenate_heads(???, num_heads)

    # Final linear projection
    output = attention_output @ W_O  # (???, seq_len, d_model)

    return output, attention_weights
end function
```

Question no 7 **Marks [20]**

A. Read the following papers and answer the questions

The Illusion of Thinking: Understanding the Strengths and Limitations of Reasoning Models via the Lens of Problem Complexity

1. What is the accuracy collapse phenomenon ?
2. What is the significance of using controllable puzzle environments instead of traditional math benchmarks?
3. Explain the “overthinking phenomenon”?
4. What central assumption about Large Reasoning Models (LRMs) does the paper challenge?
5. Why might non-thinking LLMs outperform reasoning models at low complexity? What does this suggest about the cost-benefit trade-off of chain-of-thought reasoning?
6. Should reasoning trace quality become a standard evaluation metric? What challenges would arise in implementing this?

B. Read the following papers and answer the questions

Am I Self-Conscious? (Or Does Self-Organization Entail Self-Consciousness?)

7. Can a self-aware model reduce reasoning collapse?
8. Explain the differences between Evolutionary Selection and Conscious Selection.
9. What role does counterfactual reasoning play in self-consciousness?
10. When does the transition from biological inference to conscious inference occurs? What are the main processes involved?

Good luck